CHEM 110 Exam 3 Study Guide

Built from four past Exam 3 forms (Practice A, Practice B, Form C, Form D; 100 questions). Contents: 1 Topic map · 2 Constants · 3 Equations · 4 Procedures · 5 Common mistakes · 6 Units & sig figs · 7 Practice set · 8 Answer key (separate section).

1. What the exams test most

Each past Exam 3 is 25 multiple-choice questions × 4 points = 100. A periodic table and data sheet are provided.

Topic	A	B	C	D	Total
Amino acids, peptides, nucleic acids & dihedrals	8	7	8	5	**28** (28%)
Gases: ideal gas law, KMT, effusion & real gases	7	6	6	7	**26** (26%)
Solutions, electrolytes, dilution & solubility	4	4	4	5	**17** (17%)
Phases, vapor pressure, heating curves & IMF	3	4	3	5	**15** (15%)
Calorimetry, heat capacity & solution energetics	3	4	4	3	**14** (14%)

Columns: Practice Form A, Practice Form B, Form C, Form D. Each question counted once under its main skill.

2. Constants

Gas constant	$R = 0.08206\ \text{L·atm·mol}^{-1}\text{·K}^{-1} = 8.314\ \text{J·mol}^{-1}\text{·K}^{-1}$
Avogadro's number	$N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$
Temperature	$K = °C + 273.15$
Pressure	1 atm = 760 torr = 760 mmHg
Water heat data (typical exam sheet)	$C_m(\text{ice}) = 37.6\ \text{J·mol}^{-1}\text{·K}^{-1}$; $C_m(\text{liquid}) = 75.2\ \text{J·mol}^{-1}\text{·K}^{-1}$; $\Delta H_{fus} = 6.02\ \text{kJ/mol}$
Specific heat (dilute aqueous)	$c \approx 4.184\ \text{J·g}^{-1}\text{·K}^{-1}$

3. Key equations

Ideal gas law	$PV = nRT$ T in kelvin; R = 0.08206 with P in atm and V in L.
Charles / Gay-Lussac	$V/T = \text{const}$ (P,n fixed); $P/T = \text{const}$ (V,n fixed) Always convert °C → K before ratios.
Dalton's law	$P_i = \chi_i P_{total}$ Mole fractions unchanged by compression if n fixed.
Gas density	$d = PM/RT$ Plot of d vs 1/T at fixed P has slope PM/R.
Graham's law	$\text{rate}_A/\text{rate}_B = \sqrt{(M_B/M_A)}$ Lighter gas effuses/diffuses faster.
urms	$u_{rms} = \sqrt{(3RT/M)}$ M in kg/mol when $R = 8.314\ \text{J·mol}^{-1}\text{·K}^{-1}$.

Henry's law	$c = k_H P_{gas}$ P_gas = χ · P_total for a mixture.
Molarity from mass %	Assume 1.00 L → mass = d×1000 g; mass solute = % × mass; M = (g/MolarMass) Watch ions: $Na_3PO_4 \rightarrow 3\ Na^+$.
Calorimetry	$q = m\ c\ \Delta T$ or $q = n\ C_m\ \Delta T$ Insulated: Σq ≈ 0. Include phase-change terms (nΔH) when T crosses a transition.
Heat of solution sign	$\Delta T_{solution} > 0 \Rightarrow \Delta H_{soln} < 0$ (exo); $\Delta T < 0 \Rightarrow \Delta H_{soln} > 0$ (endo) Microscopic: forming ion–dipole is exo; breaking lattice / water H-bonds is endo.

4. Step-by-step procedures

4.1 Ideal-gas temperature or pressure change

1. Convert every temperature to kelvin.
2. Identify what is held constant (V rigid → P/T; P balloon → V/T).
3. Solve the ratio; convert back to °C only at the end if asked.

4.2 Effusion / molar mass from rates

1. Write Graham: $rate_1/rate_2 = \sqrt{(M_2/M_1)}$.
2. If one gas is O_2, $M_{unk} = 32 \times (rate_{O_2}/rate_{unk})^2$.
3. Match M to a formula on the choice list.

4.3 Henry's law with air mixtures

1. Find P_gas = χ · P_total.
2. c = k_H · P_gas.
3. Convert M ↔ mM (×1000) to match the answer units.

4.4 Dilution & ion concentrations

1. mmol of ion = mL × M × (ions per formula).
2. After mixing, divide by the TOTAL volume.
3. For a target concentration, V_final = n_ion / C_target.

4.5 Mass % + density → molarity (and [ion])

1. Work with 1.00 L of solution.
2. Total grams = density (g/mL) × 1000.
3. Grams solute = mass-fraction × total grams; moles = g / M.
4. For [cation], multiply salt molarity by the subscript.

4.6 Metal / water calorimetry

1. $q_{water} = m_w \cdot 4.184 \cdot (T_f - T_w)$.
2. $q_{metal} = -q_{water} = m_{metal} \cdot c \cdot (T_f - T_{metal})$.
3. Solve for c; compare to tabulated metals.

4.7 Ice added to a warm liquid

1. Warm ice to 0°C ($n\ C_{m,ice}\ \Delta T$).
2. Melt ice ($n\ \Delta H_{fus}$).
3. Warm resulting water from 0°C to T_f ($n\ C_{m,liq}\ \Delta T$).
4. That total q came from the other liquid cooling: $q = n_{liq}\ C_{m,liq}\ (T_i - T_f)$.

4.8 Reading phase diagrams & heating curves

1. Identify the starting point (solid/liquid/gas).
2. Lowering P at fixed T often vaporizes a liquid.
3. On a heating curve: slopes ↔ C_m; plateaus ↔ ΔH_fus or ΔH_vap (usually ΔH_vap is the longer plateau).

4.9 Peptides, φ/ψ, and nucleotide mimics

1. N-terminus = free amino end (lower residue numbers); C-terminus = free carboxyl.
2. Most peptide bonds are trans; cis often involves proline.
3. φ/ψ are backbone single-bond dihedrals that set fold (Ramachandran).
4. DNA sugars lack 2′-OH; RNA has 2′-OH. Base pairs: A–T (DNA) / A–U (RNA), G–C.

5. Common mistakes

- Forgetting °C → K in any gas-law ratio (the most common Exam 3 trap).
- Using total pressure instead of partial pressure in Henry's law.
- After mixing solutions, dividing by only one volume instead of V_1+V_2 (or the stated final V).
- Counting formula units instead of ions ($CaCl_2$ gives 2 Cl^-; Na_3PO_4 gives 3 Na^+).
- Treating NH_3 or other weak bases as strong electrolytes.
- Thinking melting/vaporization releases heat — both are endothermic; condensation/freezing are exothermic.
- Mixing up which microscopic step is exo vs endo when a salt dissolves.
- Reading a solubility curve wrong: below = unsaturated, on = saturated, above without ppt = supersaturated.
- Assuming φ/ψ describe side-chain rotamers or cis/trans of the peptide bond itself.
- Forgetting that lighter gases leave a mixture faster (χ of the heavy gas rises).
- Confusing DNA vs RNA nucleotide mimics (look for 2′-OH).
- Reporting °C when the algebra was done in K (or the reverse).

6. Units and significant figures

Pressure	atm, torr
Volume	L, mL (1000 mL = 1 L)
Concentration	M = mol/L; mM = mmol/L
Energy / enthalpy	J, kJ; molar heat capacity $J\cdot mol^{-1}\cdot K^{-1}$; specific heat $J\cdot g^{-1}\cdot K^{-1}$
Temperature	K for every gas law; ΔT in °C equals ΔT in K

- Keep extra digits until the end; match the least precise measured value.
- Exact conversions (1000 mL/L, 273.15) don't cut sig figs.
- On multiple choice, units and conceptual sign errors matter more than off-by-one in the last digit.

7. Practice set

Original questions in the style of Exam 3. Multiple-choice: circle one. Calculations: show work and give units. Have a periodic table and calculator. Answers and worked solutions are in Section 8 — try every question first.

Gases: ideal gas law, KMT, effusion & real gases

1. [Easy] How many moles of ideal gas occupy 1.50 L at 3.50 atm and 50.0°C?

Answer: _____ mol

2. [Medium] In an ideal Ne/Ar mixture at STP, the denser sample is the one with

(A) higher mole fraction of Ne
(B) higher mole fraction of Ar
(C) equal Ne and Ar always the same
(D) whatever has higher T

3. [Medium] The Henry's-law constant for a gas is 1.4e-03 M/atm. What is the dissolved concentration (in mM) when the total pressure is 1.50 atm and the gas mole fraction is 0.100?

Answer: _____ mM

4. [Medium] Graham's law says the effusion rate ratio of two gases equals

(A) M1/M2
(B) √(M1/M2)
(C) √(M2/M1)
(D) (M2/M1)^2

5. [Easy] How many moles of ideal gas occupy 12.0 L at 0.800 atm and 100.0°C?

Answer: _____ mol

6. [Easy] At constant T, decreasing the volume of an ideal-gas mixture

(A) increases each partial pressure but leaves mole fractions unchanged
(B) increases mole fractions but not partial pressures
(C) changes neither P nor χ
(D) makes the gas become a liquid always

7. [Medium] Air is ~21 mol% O2. If pure O2 at 1 atm dissolves at 2.00 mM, O2 from air at 1 atm dissolves at about

(A) 2.00 mM
(B) 0.42 mM
(C) 9.5 mM
(D) 0.21 mM

8. [Medium] How many times faster does He effuse than SO2 under the same conditions? (M_SO2 = 64.06 g/mol, M_He = 4.003 g/mol)

Answer: _____

9. [Hard] The density of an ideal gas plotted versus 1/T gives a straight line of slope 800 g·K·L^{-1} at constant P = 1.00 atm. What is the molar mass of the gas?

Answer: _____ g/mol

10. [Medium] Partial pressure of gas i equals
(A) $\chi_i \times P_{total}$
(B) P_{total} / χ_i
(C) χ_i / P_{total}
(D) always 1 atm

Phases, vapor pressure, heating curves & IMF

11. [Medium] Raising pressure on a liquid near its liquid–gas curve tends to
(A) favor the gas
(B) favor the liquid
(C) always freeze the sample
(D) have no effect

12. [Medium] When ionic CsBr dissolves in water, an endothermic microscopic step is
(A) forming ion–dipole interactions
(B) forming new ionic bonds in solution
(C) breaking hydrogen bonds between water molecules
(D) forming covalent bonds to water

13. [Easy] On a phase diagram, starting in the liquid region and lowering pressure at constant T can cause
(A) freezing
(B) vaporization
(C) an increase in melting point only
(D) nothing ever

14. [Medium] The liquid expected to have the highest viscosity at a given T is the one with
(A) the weakest IMFs
(B) the strongest IMFs and/or H-bonding network
(C) the lowest boiling point always
(D) only London forces

15. [Hard] If a heating-curve plateau for vaporization is longer than the fusion plateau (same molar sample), then
(A) $\Delta H_{fus} > \Delta H_{vap}$
(B) $\Delta H_{vap} > \Delta H_{fus}$
(C) the liquid has lower Cm than the solid
(D) the gas has the lowest Cm

16. [Easy] Raising temperature at fixed composition generally makes viscosity
(A) increase
(B) decrease
(C) stay exactly the same
(D) become infinite

Solutions, electrolytes, dilution & solubility

17. [Hard] Which 1.0 mM solution has the lowest total ion concentration?
(A) 1.0 mM NH4Br (strong electrolyte)
(B) 1.0 mM CH3Br (nonelectrolyte)
(C) 1.0 mM Na2SO4
(D) 1.0 mM CaCl2

18. [Hard] To what volume (L) must you dilute 250 mL of 0.500 M $CaCl_2$ to obtain a solution that is 150 mM in Cl^-? ($CaCl_2 \rightarrow 2\ Cl^-$)
Answer: _____ L

19. [Hard] On a solubility-vs-T curve, a point below the curve represents
(A) a saturated solution
(B) an unsaturated solution
(C) a supersaturated solution
(D) a pure solid solute

20. [Hard] A 4.00% by mass aqueous Na_2SO_4 solution has density 1.03 g/mL. What is the concentration of the cation in mM?
Answer: _____ mM

21. [Medium] Among 10 mM FeCl3, 30 mM HClO3, and 20 mM HCl, the greatest [Cl−] is from
(A) 20 mM HCl
(B) 30 mM HClO3
(C) 10 mM FeCl3
(D) they are all equal at 10 mM

22. [Hard] To what volume (L) must you dilute 100 mL of 0.800 M $CaCl_2$ to obtain a solution that is 100 mM in Cl^-? ($CaCl_2 \rightarrow 3\ Cl^-$)
Answer: _____ L

23. [Medium] A point on a solubility curve is a solution that is
(A) unsaturated
(B) saturated
(C) supersaturated
(D) only gaseous

Calorimetry, heat capacity & solution energetics

24. [Medium] Forming ion–dipole contacts during hydration is
(A) endothermic
(B) exothermic
(C) impossible for salts
(D) only for covalent solids

25. [Easy] In an insulated calorimeter, heat lost by hot metal equals
(A) zero
(B) heat gained by the water (approx.)
(C) water's boiling point
(D) nR

26. [Easy] Cooling 1 mol of a substance with a large Cm by 10°C releases
(A) less heat than cooling a low-Cm substance by 10°C
(B) more heat than cooling a low-Cm substance by 10°C
(C) the same heat as melting it
(D) zero heat

27. [Hard] A 40.0 g mixture of ice and water at 0°C absorbs 10.0 kJ of heat and reaches 20.0°C. How many grams of ice were in the original mixture? (ΔH_{fus}=6.02 kJ/mol, C_m water=75.2 J·mol^{-1}·K^{-1})
Answer: _____ g

28. [Medium] Dissolving CsBr has ΔHsoln = +26 kJ/mol. When CsBr dissolves, the solution temperature
(A) increases
(B) decreases
(C) stays exactly the same
(D) must boil

29. [Medium] Specific heat capacity has units of
(A) J only
(B) J·g−1·K−1
(C) K only
(D) mol only

Amino acids, peptides, nucleic acids & dihedrals

30. [Medium] DNA base pairing pairs adenine with
(A) guanine
(B) cytosine
(C) thymine
(D) uracil

31. [Medium] A Ramachandran plot maps
(A) pKa vs pH for side chains
(B) allowed φ,ψ angle combinations
(C) hydrophobicity vs residue number
(D) cis vs trans peptide bond frequency only

32. [Medium] When methane carbon goes from CH4 to CO2 it is
(A) reduced
(B) oxidized
(C) unchanged
(D) polymerized

33. [Hard] Which change to a nucleoside sugar would most clearly mark a drug as DNA-like rather than RNA-like?
(A) Adding a 2′-OH
(B) Removing the 2′-OH (2′-deoxy)
(C) Adding a 3′-phosphate
(D) Replacing the base with adenine

34. [Medium] The peptide bond is typically
(A) freely rotating like ethane
(B) cis for every AA
(C) planar with a strong trans preference
(D) a triple bond

35. [Easy] In $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$, carbon is
(A) oxidized
(B) reduced
(C) unchanged in oxidation state
(D) a catalyst

36. [Medium] An antiviral nucleotide mimic with a 2′-OH on the sugar is designed to resemble
(A) a DNA nucleotide
(B) an RNA nucleotide
(C) an amino acid
(D) a fatty acid

37. [Easy] The φ (phi) and ψ (psi) dihedral angles of a polypeptide primarily describe
(A) side-chain rotamers
(B) the folded shape of the backbone
(C) whether a residue is D or L
(D) the cis/trans state of every peptide bond

38. [Easy] A vitamin structure least soluble in water is typically
(A) rich in OH groups
(B) a large nonpolar hydrocarbon-rich molecule
(C) a charged phosphate ester
(D) heavily H-bonding

39. [Medium] Dihedral angles such as φ and ψ measure
(A) the bond angle at Cα
(B) rotation about a single bond
(C) the length of a peptide bond
(D) hydrogen-bond strength

40. [Medium] A cis peptide bond is most often found involving which amino acid?
(A) Glycine
(B) Proline
(C) Alanine
(D) Serine

8. Answer key with worked solutions

Separate section — fold or detach before working the practice set.

1. Answer: 0.198 mol

1. T = 50.0 + 273.15 = 323.15 K
2. n = PV/RT = (3.50)(1.50)/(0.08206 × 323.15)
3. n = **0.198 mol**

2. Answer: (B) higher mole fraction of Ar

1. d = PM/RT; at same P,T denser means larger M.
2. Higher χ_Ar raises M_avg.

3. Answer: 0.210 mM

1. P_gas = 0.100 × 1.50 = 0.1500 atm
2. c = (1.4e-03 M/atm)·(0.1500 atm) = 2.1000e-04 M
3. = **0.210 mM**

4. Answer: (C) √(M2/M1)

1. rate_A/rate_B = √(M_B/M_A).

5. Answer: 0.314 mol

1. T = 100.0 + 273.15 = 373.15 K
2. n = PV/RT = (0.800)(12.0)/(0.08206 × 373.15)
3. n = **0.314 mol**

6. Answer: (A) increases each partial pressure but leaves mole fractions unchanged

1. Compressing raises total P.
2. Mole fractions depend only on mole amounts.

7. Answer: (B) 0.42 mM

1. c = k P_O2 = 2.00 mM/atm × 0.21 atm ≈ 0.42 mM.

8. Answer: 4.00

1. rate(He)/rate(SO2) = √(M_SO2/M_He) = √(64.06/4.003)
2. = **4.00**

9. Answer: 65.6 g/mol

1. slope = PM/R → M = slope·R/P = 800×0.08206/1.00
2. M = **65.6 g/mol**

10. Answer: (A) χ_i × P_total

1. P_i = χ_i P_tot.

11. Answer: (B) favor the liquid

1. Compression can condense vapor.

12. Answer: (C) breaking hydrogen bonds between water molecules

1. Separating water molecules (breaking H-bonds) is endothermic.
2. Forming ion–dipole is exothermic.

13. Answer: (B) vaporization

1. Crossing the liquid–gas curve by lowering P → vaporization.

14. Answer: (B) the strongest IMFs and/or H-bonding network

1. H-bonding liquids (e.g. acetic acid, glycerol) are more viscous than acetone-like solvents.

15. Answer: (B) ΔHvap > ΔHfus

1. Compare lengths of the two flat regions.

16. Answer: (B) decrease

1. Higher T → lower viscosity for liquids.

17. Answer: (B) 1.0 mM CH3Br (nonelectrolyte)

1. CH3Br does not ionize. Salts give multiple ions.

18. Answer: 1.67 L

1. n(Cl^-) = 0.250 L × 0.500 mol/L × 2 = 0.250 mol
2. V = 0.250 / (0.150) = **1.67 L**

19. Answer: (B) an unsaturated solution

1. On the curve = saturated; above = supersaturated.

20. Answer: 870 mM

1. M_salt = (0.04×1030)/142.04 = 0.2901 M
2. [cation] = 0.2901 × 3 = 0.8702 M = **870 mM**

21. Answer: (C) 10 mM FeCl3

1. 10 mM FeCl3 → 30 mM Cl−. HClO3 gives ClO3− not Cl−. HCl → 20 mM Cl−.

22. Answer: 2.40 L

1. n(Cl^-) = 0.100 L × 0.800 mol/L × 3 = 0.240 mol
2. V = 0.240 / (0.100) = **2.40 L**

23. Answer: (B) saturated

1. Saturated: dissolved ⇌ solid.

24. Answer: (B) exothermic

1. Hydration enthalpy is negative.

25. Answer: (B) heat gained by the water (approx.)

1. Isolated cup ≈ adiabatic.

26. Answer: (B) more heat than cooling a low-Cm substance by 10°C

1. Larger Cm → larger |q| for the same ΔT.

27. Answer: 19.9 g

1. Heat to warm all water 0→20.0: (40.0/18.015)·75.2·20.0 = 3340 J
2. Heat left to melt ice: 10000 − 3340 = 6660 J
3. m_ice = q_melt / (ΔH_{fus}/M) = **19.9 g**

28. Answer: (B) decreases

1. ΔH > 0 → system takes heat → solution cools.

29. Answer: (B) J·g−1·K−1

1. Energy per mass per degree.

30. Answer: (C) thymine

1. Watson–Crick pairs: A with T (two H-bonds), G with C (three).

31. Answer: (B) allowed φ,ψ angle combinations

1. Steric clashes forbid many φ,ψ pairs.
2. Allowed islands correspond to α-helix, β-sheet, etc.

32. Answer: (B) oxidized

1. Increase in oxidation number = oxidation.

33. Answer: (B) Removing the 2′-OH (2′-deoxy)

1. RNA nucleotides retain the 2′-OH.
2. 2′-deoxy sugars are the DNA hallmark.

34. Answer: (C) planar with a strong trans preference

1. Resonance → planar; trans preferred.

35. Answer: (A) oxidized

1. Oxidation number of C increases → oxidation.

36. Answer: (B) an RNA nucleotide

1. Ribose (RNA) has OH at C2′; deoxyribose (DNA) does not.
2. A 2′-OH mimic targets RNA-like substrates.

37. Answer: (B) the folded shape of the backbone

1. φ is rotation about N–Cα; ψ about Cα–C.
2. Together they define secondary structure / fold.

38. Answer: (B) a large nonpolar hydrocarbon-rich molecule

1. Nonpolar molecules dislike water.

39. Answer: (B) rotation about a single bond

1. A dihedral is the angle between two planes sharing a bond.
2. For peptides, that bond is a backbone single bond.

40. Answer: (B) Proline

1. Most peptide bonds are trans.
2. Proline's cyclic side chain stabilizes cis conformations more than other AAs.